The Effect of Feedback Modality on Speed–Accuracy Tradeoff in a Visual Target-Selection Task

MIE 286 — Probability & Statistics (Winter 2026)

Final Report

Team 51

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1 Introduction

Real-time performance feedback is a core component of interactive systems, from factory control panels to surgical interfaces to video games. In each case, users must respond quickly while maintaining accuracy, and the form that feedback takes—visual, auditory, or haptic—may shape how they navigate this speed–accuracy tradeoff (SAT). Wickens' Multiple Resource Theory (MRT) predicts that pairing a visual task with visual feedback should overload a single attentional channel, whereas auditory feedback distributes processing across modalities and reduces interference [1, 2]. Under this model, auditory feedback should yield faster, less disrupted responses.

We tested this prediction in a browser-based target-selection task and found a surprising reversal: participants were *faster* under visual feedback, not slower. The visual condition used a gold/green/red border glow that created a salient speed incentive—participants could see whether they were "fast" or "slow"—and this drove speed-chasing behavior that overrode MRT's predicted interference. This reversal, and what it reveals about the motivational dimension of feedback, is the central finding of this report.

We investigated three hypotheses:

1. **H_1 (Speed):** Mean reaction time will differ between visual and auditory feedback conditions.
2. **H_2 (Accuracy):** The faster condition will exhibit greater radial error, consistent with a speed–accuracy tradeoff.
3. **H_3 (SAT structure):** The relationship between reaction time and radial error will differ between conditions—the SAT slope will be steeper under one modality.

2 Background

Wickens' Multiple Resource Theory posits that human attention operates through distinct pools organized by modality, processing stage, and response code [1, 2]. A visual pointing task paired with visual feedback should compete for the same attentional channel, whereas auditory feedback draws on a separate pool and should reduce interference. Cognitive Load Theory similarly predicts that cross-modal distribution of information lowers cognitive burden [3].

Motor performance in pointing tasks is well characterized by Fitts' Law, which relates movement time to an index of difficulty defined by target distance and width [4, 5]. Our task holds target size constant and randomizes position, so performance differences should reflect cognitive processing of feedback rather than motor constraints alone. Consistent with Fitts' Law, we observed that edge-placed targets (requiring longer cursor travel) took 64 ms longer

than center targets (753 ms vs. 689 ms).

The speed–accuracy tradeoff is a fundamental constraint: faster responses are generally less accurate. Heitz [6] emphasizes that the SAT is not a single parameter but a *function* whose shape can change across conditions. We ask whether feedback modality changes this function. If visual feedback creates speed pressure, the SAT slope under visual should be steeper—each millisecond of haste costs more accuracy.

Prior work supports modality effects. Akamatsu et al. [7] found that auditory feedback reduced movement time in a Fitts' Law task, and Sigrist et al. [8] reviewed evidence that auditory feedback supports motor learning through cross-modal facilitation. However, these studies examined mean performance rather than the tradeoff structure. Whether feedback modality changes the SAT slope remains an open question.

3 Methods

3.1 Participants

Fifty-five individuals (ages 18–24, university students) participated via convenience sampling. Covariates recorded: input device (Mouse, $n = 17$; Trackpad, $n = 27$; Other, $n = 11$), gaming frequency (Never, $n = 12$; Rarely, $n = 19$; Weekly, $n = 19$; Daily, $n = 5$), and condition order (Visual-first, $n = 27$; Auditory-first, $n = 28$).

3.2 Apparatus and Task

The experiment was a browser-based application hosted on GitHub Pages (screenshots in Appendix C). The task canvas measured 800×500 px. Circular targets (radius = 35 px) appeared at random positions, and participants clicked on them as quickly and accurately as possible. Timing used the browser's `performance.now()` API (sub-millisecond resolution). A 400 ms inter-trial interval separated targets.

3.3 Independent Variable

Feedback type was a within-subject variable with two levels:

- **Visual:** A border glow—gold for fast hits (<600 ms), green for slow hits (≥600 ms), and red with a shake animation for misses. Duration: 150 ms.
- **Auditory:** Pure tones—880 Hz for fast hits, 440 Hz for slow hits, and 180 Hz for misses. Duration: 200 ms.

Both conditions conveyed *identical three-level information* (fast hit, slow hit, miss). The on-screen HUD—trial counter, score, and streak—was identical in both conditions. This design addresses feedback from the proposal review, which noted an information asymmetry between conditions.

3.4 Dependent Variables

- **DV_1: Reaction time** (ms)—target onset to click.
- **DV_2: Radial error** (px)—Euclidean distance from click to target center. A hit was defined as error ≤ 35 px.

3.5 Procedure and Design

The study used a within-subject, counterbalanced design. After providing consent, participants completed: Instructions → 5 practice trials → Block 1 (30 or 60 trials) → Break → Block 2 → End screen. Condition order was randomized. Target size, canvas, ITI, and HUD were constant across conditions; target positions were randomized. A gamification system (scoring, particle effects, and a leaderboard with superlatives) was introduced mid-collection to boost engagement and recruitment.

3.6 Potential Biases and Limitations

Convenience sampling limits generalizability. Device variability introduced noise (mouse users were 142 ms faster than trackpad users). Trial counts varied: 30 participants completed 30 trials/block and 25 completed 60; per-participant means normalize this. Audio environment was uncontrolled. The leaderboard introduced mid-collection may have shifted motivation: post-leaderboard participants were 77 ms faster on average ($p = .007$), though accuracy was unchanged. Four participants completed the experiment more than once with different settings; each attempt was treated as a separate observation.

4 Analysis

4.1 Data Cleaning and Aggregation

Trials with RT < 150 ms (anticipatory) or $> 8{,}000$ ms (inattention) were excluded—3 of 4,800 trials (0.06%). The remaining 4,797 trials were aggregated to per-participant means for each condition, yielding 55 paired observations.

4.2 Assumption Checks

Normality of within-subject differences was assessed with the Shapiro–Wilk test. RT differences were approximately normal ($W = 0.985$, $p = .704$), so a paired t-test was appropriate for H_1. Error differences violated normality ($W = 0.892$, $p < .001$), so the Wilcoxon signed-rank test was used for H_2. QQ plots are in Appendix A.

4.3 Hypothesis Tests

H_1 (Speed). Our original prediction, based on MRT, was that auditory feedback would produce faster responses by reducing intra-channel interference. The data showed the opposite—visual RT was numerically lower. We used a *two-tailed* paired t-test because the observed direction contradicted our directional prediction:

$$H_0 : \mu_{\text{visual}} = \mu_{\text{auditory}}$$
$$H_1 : \mu_{\text{visual}} \neq \mu_{\text{auditory}}$$

H_2 (Accuracy). If visual feedback drives faster responses, the SAT predicts those responses should be less accurate. We used a *one-tailed* Wilcoxon signed-rank test (normality violated):

$$H_0 : \text{median}(d_{\text{error}}) \leq 0$$
$$H_1 : \text{median}(d_{\text{error}}) > 0, \quad d_{\text{error}} = \bar{e}_{\text{visual}} - \bar{e}_{\text{auditory}}$$

H_3 (SAT structure). We fit a multiple linear regression with an interaction term at the trial level:

$$\text{error} = \beta_0 + \beta_1 \cdot \text{RT} + \beta_2 \cdot \text{condition} + \beta_3 \cdot (\text{RT} \times \text{condition}) + \varepsilon$$

A significant β_3 indicates different SAT slopes between conditions. We supplemented this with Pearson correlations and simple linear regressions per condition.

All tests used $\alpha = 0.05$. Cohen's d is reported for paired comparisons. With $n = 55$ and the observed small effect sizes ($d \approx 0.15$), the study is underpowered for H_1 and H_2 (power ≈ 0.25), increasing the probability of Type II error.

5 Results

5.1 Descriptive Statistics

Table 1 summarizes performance by condition. Visual feedback was associated with numerically faster responses (705 vs. 715 ms) and slightly greater radial error (15.3 vs. 14.6 px). Hit rates were nearly identical (∼95%). The pattern is consistent with a speed–accuracy tradeoff under visual feedback, though the raw differences are small.

Table 1: Descriptive statistics by feedback condition ($N = 55$). Values are means of per-participant means ± between-participant SD.

	Visual	**Auditory**
Reaction Time, $M \pm SD$ (ms)	705 ± 111	715 ± 115
Radial Error, $M \pm SD$ (px)	15.3 ± 5.4	14.6 ± 4.4
Hit Rate (%)	95.3 ± 5.6	95.4 ± 5.1

Figure 1 shows individual participant trajectories between conditions. While the grand means differ by only 9.4 ms, individual differences are substantial—some participants were over 100 ms faster under visual feedback, others over 100 ms slower. This between-participant variability ($SD = 59$ ms for the paired differences) is the primary reason H_1 did not reach significance.

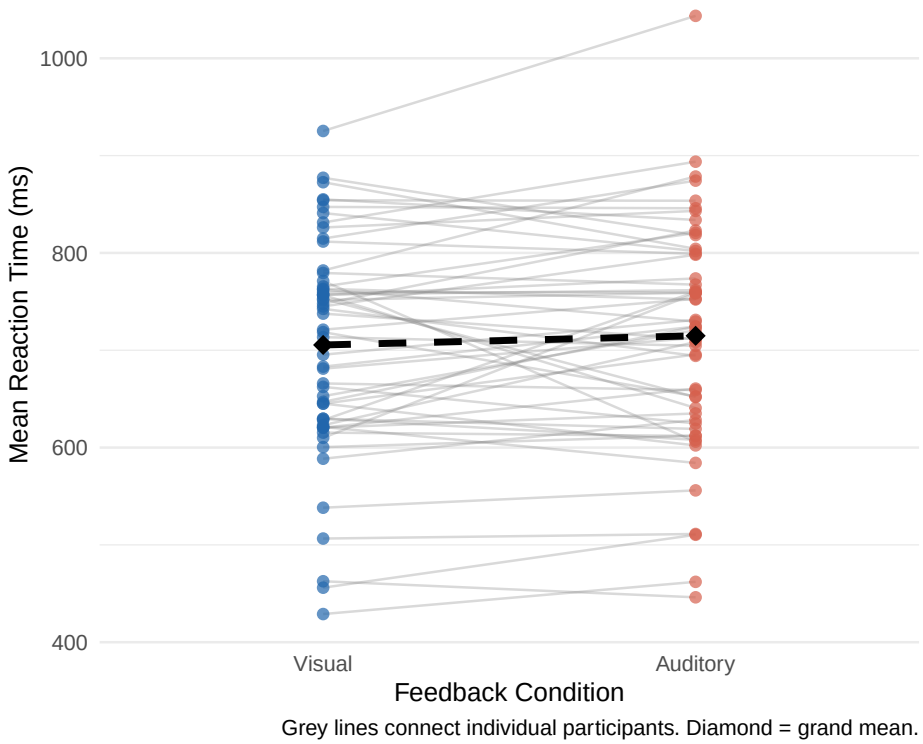


Figure 1: Paired comparison of mean RT by condition. Each grey line connects one participant's visual and auditory means. The dashed line connects grand means. Slopes vary widely across individuals.

5.2 Reaction Time Distributions

Figure 2 overlays the trial-level RT distributions. Both conditions show the characteristic right-skewed shape of reaction time data, but the visual distribution has a sharper peak and heavier concentration around 500–600 ms—the range that triggers gold feedback. This is consistent with participants actively targeting the gold threshold under visual feedback.

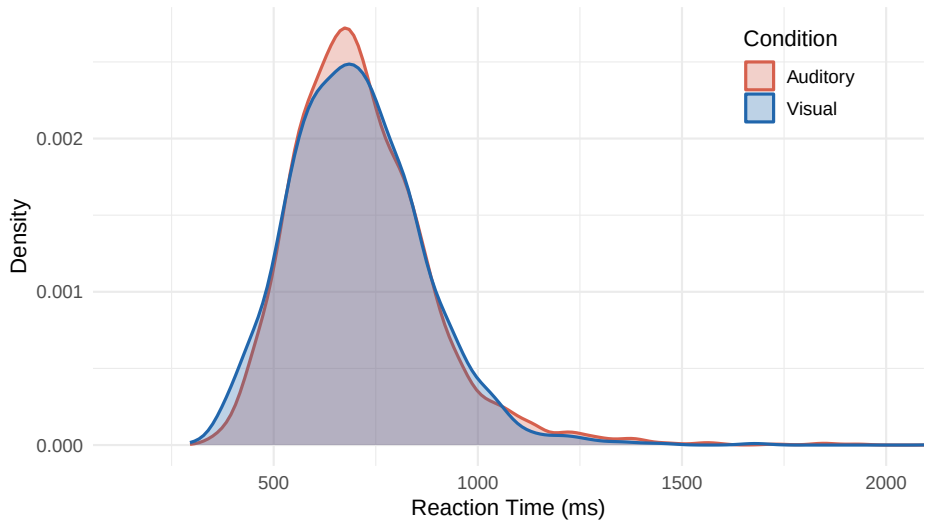


Figure 2: Trial-level RT distributions by condition. The visual condition shows a tighter, more peaked distribution concentrated near the 600 ms gold/green threshold, suggesting participants targeted this boundary.

5.3 Inferential Statistics

H_1 (Speed). A paired t-test found no significant difference in mean RT, $t(54) = -1.19$, $p = .240$, 95% CI $[-25.3,\ 6.5]$ ms, $d = -0.16$. We fail to reject H_0. The 9.4 ms difference, while in the direction opposite to MRT's prediction, was not statistically reliable given high between-participant variability.

H_2 (Accuracy). A Wilcoxon signed-rank test found no significant difference in radial error, $V = 880$, $p = .180$ (one-tailed), $d = 0.15$. Visual error was 0.69 px higher on average (median difference: 0.12 px), but this was not significant. Accuracy was near ceiling ($\sim$95% hit rate), limiting the range of detectable error differences.

H_3 (SAT structure). A multiple linear regression revealed a *significant* RT$\times$condition interaction, $\beta_3 = -0.0083$, $t(4793) = -3.73$, $p < .001$, indicating that the SAT slope was steeper under visual feedback. The full model: $F(3, 4793) = 30.47$, $p < .001$, $R^2 = .019$. Simple regressions confirmed: the visual slope ($b = -0.014$, $R^2 = .018$) was more than twice as steep as the auditory slope ($b = -0.006$, $R^2 = .016$). Pearson correlations were $r = -.135$ ($p < .001$) for visual and $r = -.126$ ($p < .001$) for auditory. Figure 3 visualizes this difference.

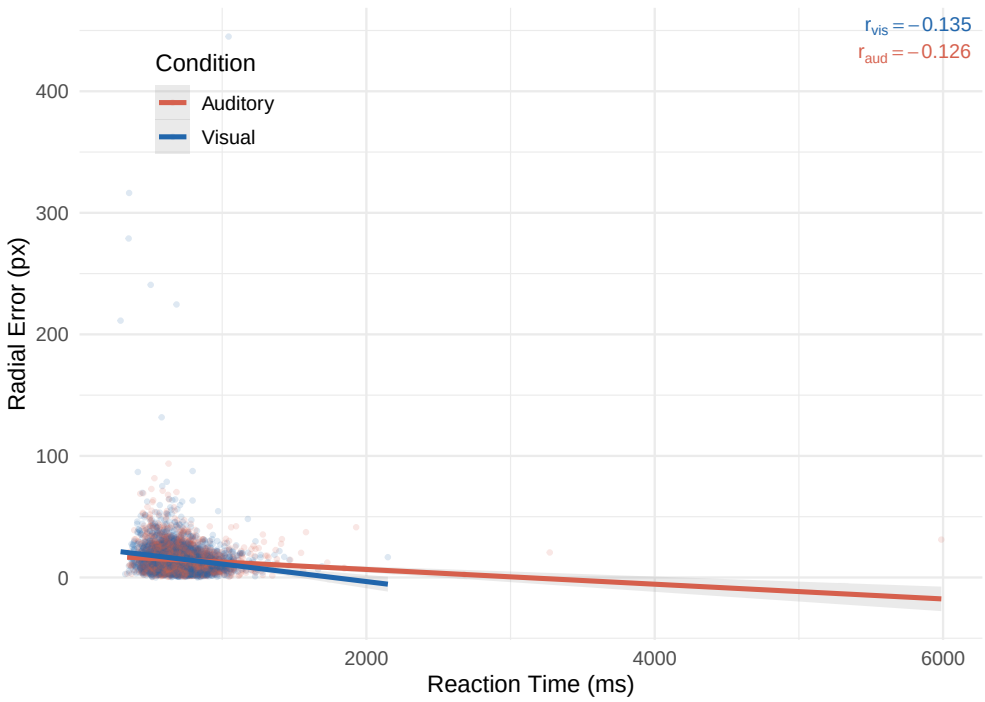


Figure 3: Speed–accuracy tradeoff by condition. Each point is one trial. The steeper regression line under visual feedback shows that faster visual responses incurred a larger accuracy penalty (interaction $p < .001$). Shaded bands show 95% confidence intervals.

5.4 Learning Effects

Figure 4 shows mean RT across trial number within a block. Both conditions show a rapid improvement in the first ∼5 trials followed by a plateau, with a slight increase in the final trials suggesting fatigue. The visual condition is consistently faster throughout, and the two curves are approximately parallel—the modality effect is stable across the block rather than emerging through practice.

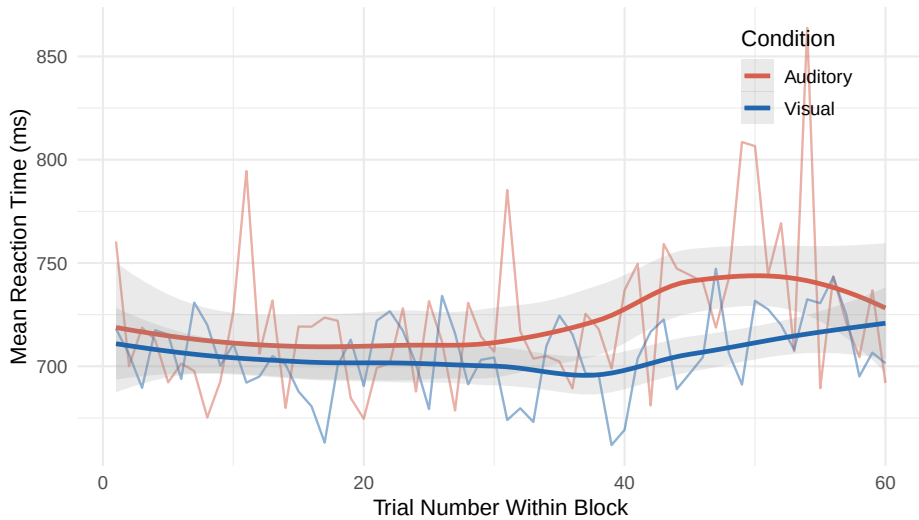


Figure 4: Mean RT by trial number within each block. Both conditions show early improvement (trials 1–5) and late fatigue (final trials). The visual advantage is stable across the block.

5.5 Spatial Click Patterns

Plotting all clicks relative to target center (Appendix B, Figure B.6) revealed a systematic downward bias: on misses, participants clicked an average of 7.9 px below the target and 2.8 px to the left. This may reflect cursor hotspot offset or hand biomechanics. The miss cloud was slightly wider under visual feedback, consistent with higher error variance when speed-chasing.

5.6 Secondary Findings

Tilt effect. The probability of a miss following a previous miss was 12.2% (26/213), compared with 4.2% (187/4,474) following a hit—nearly three times higher, $\chi^2(1) = 28.38$, $p < .001$. This "tilt" effect, analogous to the poker term for emotionally-driven errors after a loss, suggests that errors temporarily disrupt attentional focus or motor calibration, creating cascading misses.

Counterbalancing check. An independent-samples t-test comparing overall mean RT between Visual-first ($M = 707$ ms) and Auditory-first ($M = 713$ ms) groups found no significant difference, $t(52.3) = 0.19$, $p = .850$, confirming that counterbalancing was effective.

Covariates. Mouse users were substantially faster than trackpad users (629 ms vs. 771 ms), reflecting the ergonomic advantage of a dedicated pointing device (Table 2). Daily gamers were the slowest group (770 ms) but achieved the highest hit rate (97.3%), suggesting

a deliberate accuracy-over-speed strategy—a natural SAT operating at the individual level.

Table 2: Mean reaction time and hit rate by input device and gaming frequency.

	N	Mean RT (ms)	Hit Rate (%)
Device Type			
Mouse	17	629	94.2
Trackpad	27	771	95.8
Other	11	687	95.8
Gaming Frequency			
Never	12	705	95.8
Rarely	19	737	95.0
Weekly	19	671	94.8
Daily	5	770	97.3

6 Conclusion

Neither mean RT nor mean error differed significantly between conditions (H_1: $p = .240$; H_2: $p = .180$). However, the *structure* of the speed–accuracy tradeoff did differ significantly (H_3: $p < .001$): the SAT slope under visual feedback was more than twice as steep as under auditory feedback.

The MRT prediction—that auditory feedback should reduce interference and produce faster responses—was not supported. Instead, the visual condition's gold/green border glow created a salient speed incentive: participants could *see* whether they were "fast" (gold) or "slow" (green), and during data collection they were visibly frustrated at receiving green instead of gold. This speed-chasing behavior was absent in the auditory condition, where the tone distinction was less motivationally compelling. MRT's interference framework is therefore incomplete for predicting feedback effects: modality affects not only *where* attentional resources are allocated but *how aggressively* participants pursue speed.

The tilt effect ($\chi^2 = 28.38$, $p < .001$) adds further evidence: errors cascade, with a miss tripling the probability of a subsequent miss. This is consistent with attentional disruption or frustration-driven strategy shifts, not simple motor noise.

Practical implications. Visual feedback in safety-critical interfaces may inadvertently create speed pressure; auditory feedback may better preserve accuracy by decoupling the speed incentive from the visual task channel. Conversely, visual feedback may be preferred where speed is desirable.

Limitations. The convenience sample limits generalizability. Device variability introduced noise (mouse users 142 ms faster than trackpad). Mixed trial counts were addressed via per-participant means. The leaderboard ($p = .007$) may have shifted motivation but was orthogonal to the within-subject comparison. Accuracy was near ceiling ($\sim$95%), limiting detectable error differences. The visual border glow is inherently more salient than a tone; our results speak to feedback "as deployed" rather than modality *per se*.

Lessons learned. Counterbalancing was essential ($p = .850$ confirmed no order effect). Trial independence cannot be assumed in sequential motor tasks. The most informative finding was the one we did not predict—future work should manipulate feedback salience independently of modality to disentangle the motivational and interference channels.

References

[1] C. D. Wickens, "Multiple resources and performance prediction," *Theor. Issues Ergon. Sci.*, 3(2), 159–177, 2002.

[2] C. D. Wickens, J. G. Hollands, S. Banbury, R. Parasuraman, *Eng. Psychology & Human Performance*, 4th ed., Routledge, 2015.

[3] J. Sweller, "Cognitive load during problem solving: Effects on learning," *Cognitive Science*, 12(2), 257–285, 1988.

[4] P. M. Fitts, "The information capacity of the human motor system in controlling the amplitude of movement," *J. Exp. Psychol.*, 47(6), 381–391, 1954.

[5] I. S. MacKenzie, "Fitts' law as a research and design tool in human–computer interaction," *Hum.–Comput. Interact.*, 7(1), 91–139, 1992.

[6] R. P. Heitz, "The speed–accuracy tradeoff: History, physiology, methodology, and behavior," *Front. Neurosci.*, 8, 150, 2014.

[7] M. Akamatsu, I. S. MacKenzie, T. Hasbroucq, "A comparison of tactile, auditory, and visual feedback in a pointing task using a mouse-type device," *Ergonomics*, 38(4), 816–827, 1995.

[8] R. Sigrist, G. Rauter, R. Riener, P. Wolf, "Augmented visual, auditory, haptic, and multimodal feedback in motor learning: A review," *Psychon. Bull. & Rev.*, 20(1), 21–53, 2013.

A Assumption Checks

Normality of within-subject differences was assessed using the Shapiro–Wilk test and QQ plots.

- **RT differences:** $W = 0.985$, $p = .704$ — normality not rejected. Paired t-test appropriate.
- **Error differences:** $W = 0.892$, $p < .001$ — normality rejected. Wilcoxon signed-rank test used.

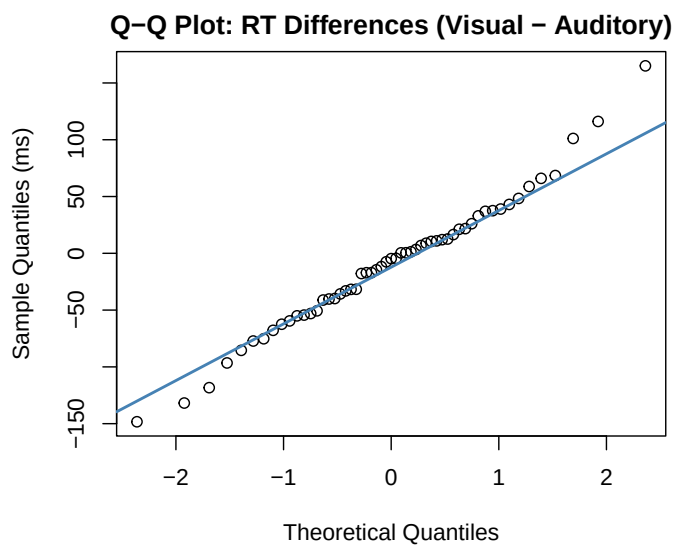


Figure 5: QQ plot for RT differences (visual − auditory). Points follow the reference line closely.

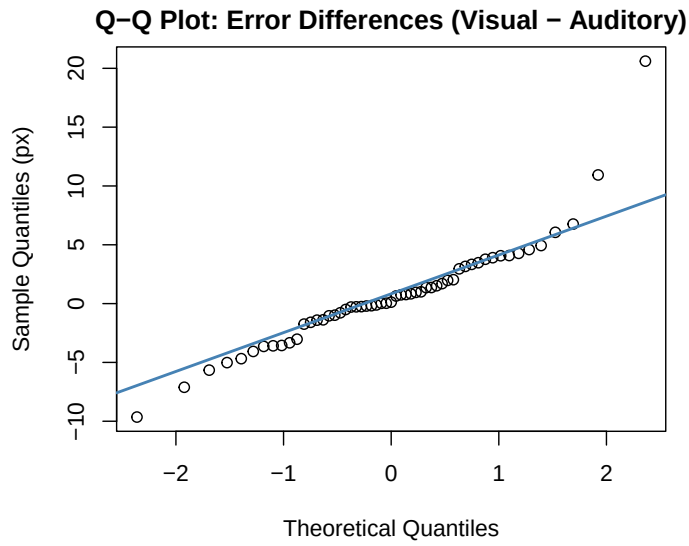


Figure 6: QQ plot for error differences (visual − auditory). Heavy tails indicate non-normality ($p < .001$).

Regression Diagnostics

Residual analysis for the SAT interaction model (error $\sim$ RT $\times$ condition):

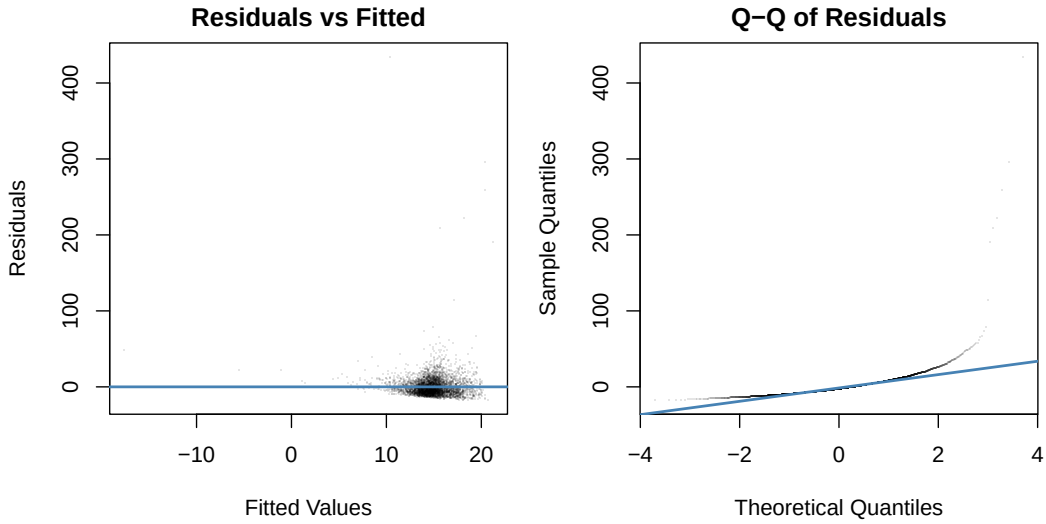


Figure 7: Residual diagnostics for the SAT regression model. Left: residuals vs. fitted values show a funnel pattern (heteroscedasticity expected with count-bounded error). Right: QQ plot of residuals shows heavy right tail from large misses.

B Additional Figures

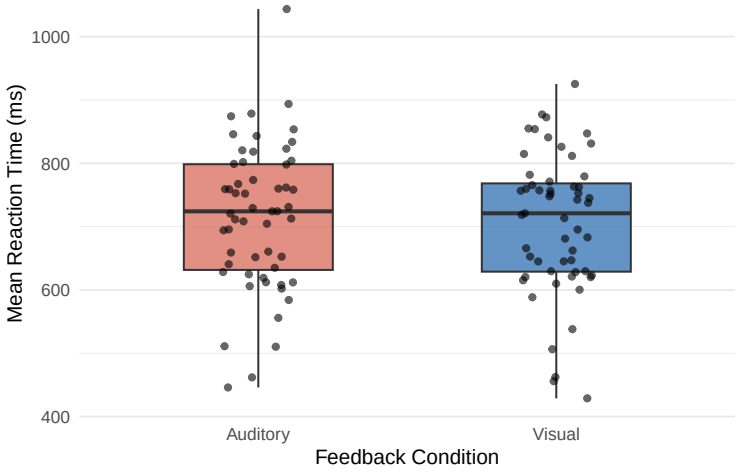


Figure 8: RT boxplots by condition with individual participant points.

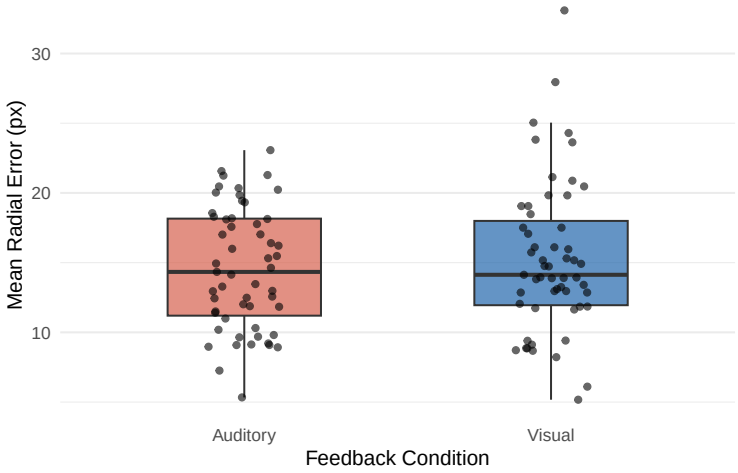


Figure 9: Radial error boxplots by condition.

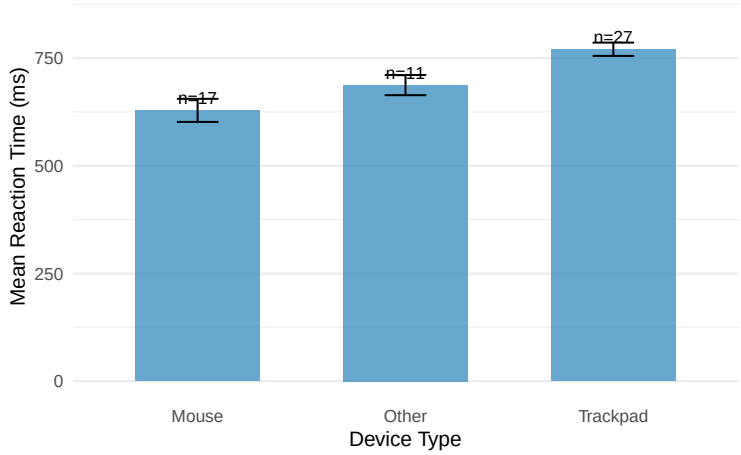


Figure 10: Mean RT by device type (±1 SE). Mouse users were 142 ms faster than trackpad users.

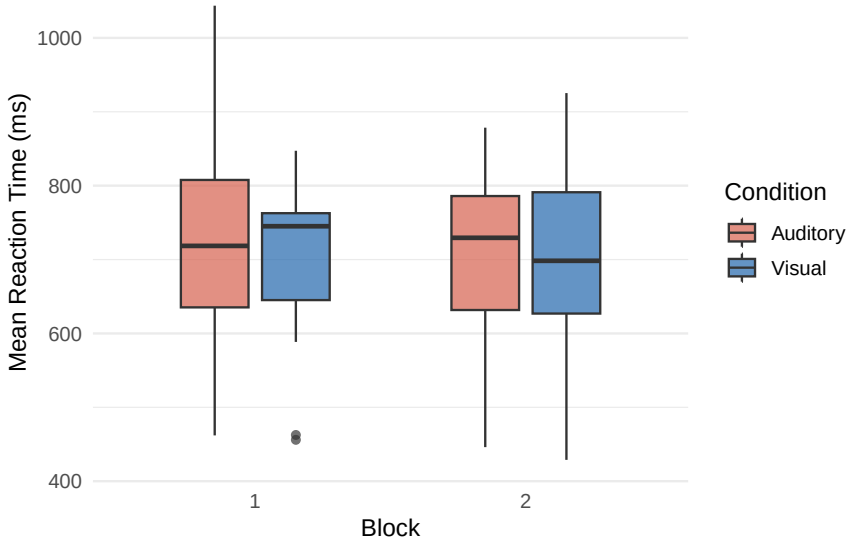


Figure 11: RT by block and condition. Block 2 was slightly faster ($M = 708$ vs. 713 ms, $p = .555$, $d = 0.08$), suggesting a small but non-significant practice effect.

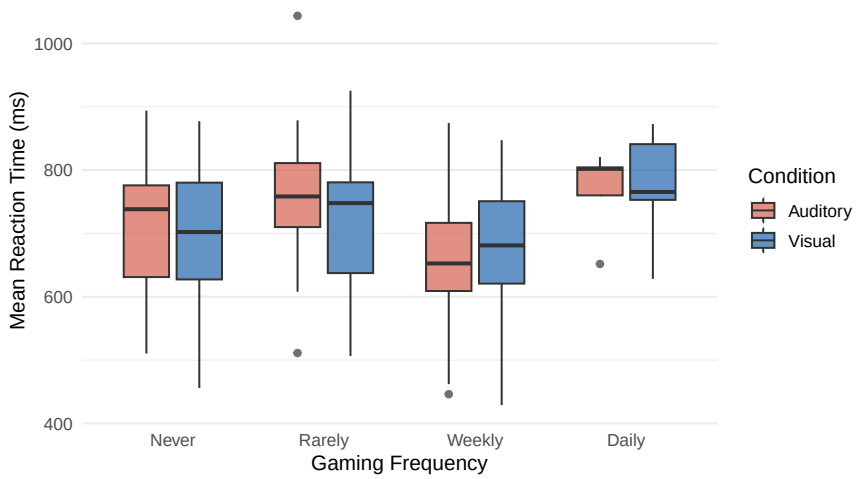


Figure 12: RT by gaming frequency and condition. Daily gamers were the slowest group but achieved the highest accuracy (97.3%), reflecting a cautious strategy.

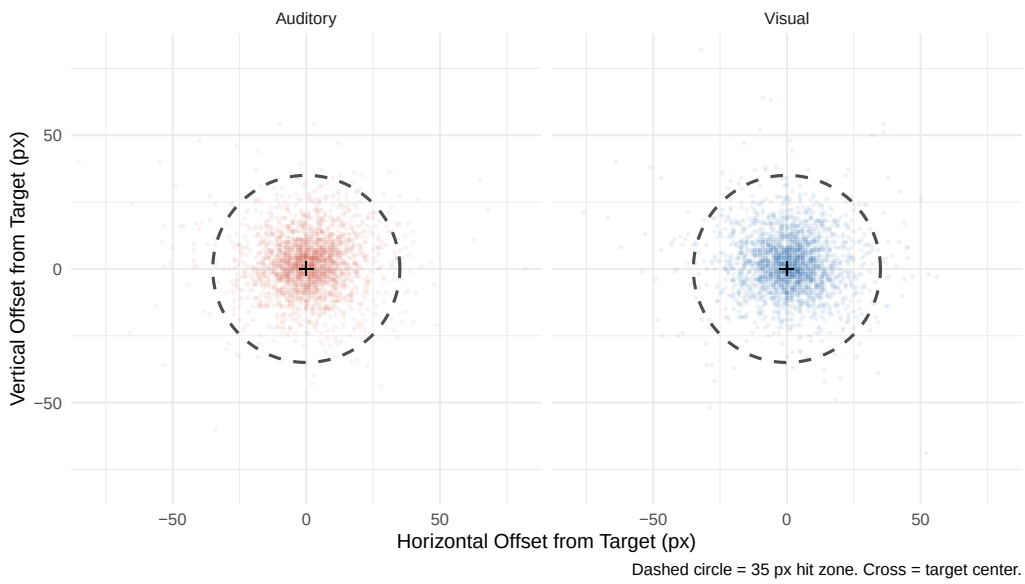
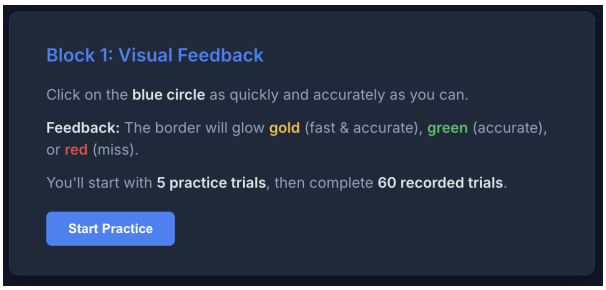
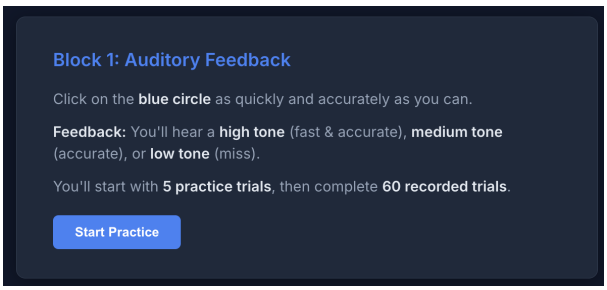


Figure 13: Click positions relative to target center by condition. Dashed circle = 35 px hit zone. Note the systematic downward bias and the wider scatter under visual feedback.

C Experiment Interface and Screenshots

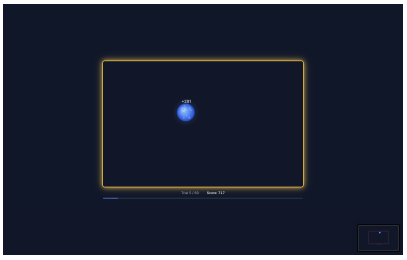


(a) Visual condition instructions

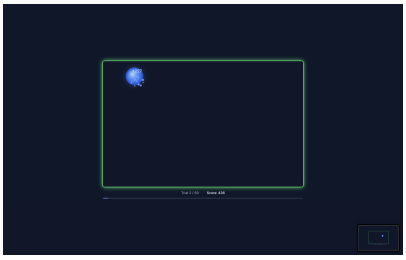


(b) Auditory condition instructions

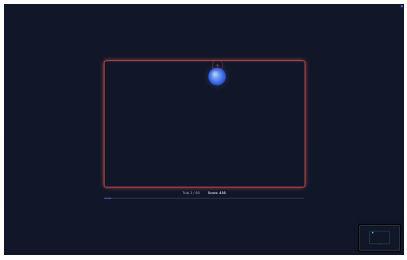
Figure 14: Instruction screens shown before each block. Both describe the same task; only the feedback description differs. Participants were told the border would glow gold/green/red (visual) or that they would hear high/medium/low tones (auditory).



(a) Gold glow (fast hit, <600 ms)

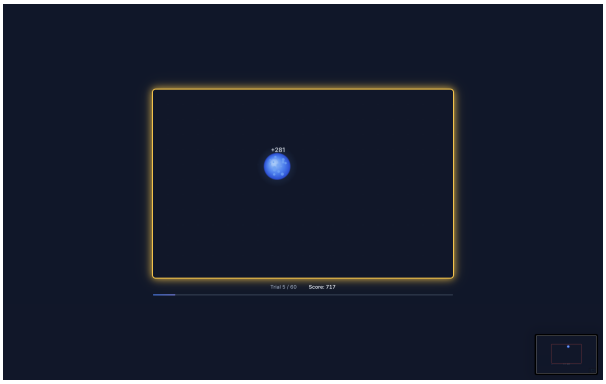


(b) Green glow (slow hit, ≥600 ms)

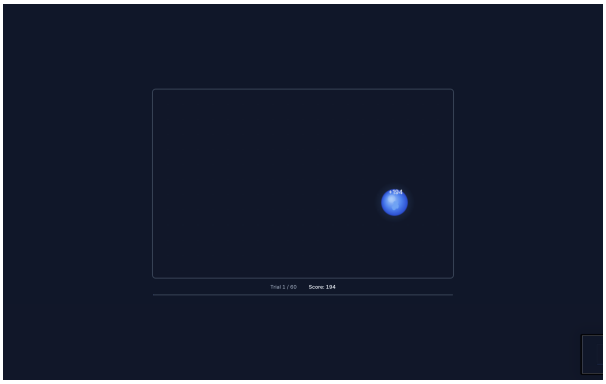


(c) Red glow + shake (miss)

Figure 15: The three visual feedback states. The salient distinction between gold and green—visible at a glance—created the speed incentive that drove participants to chase faster responses. Participants were observed to be visibly frustrated at receiving green instead of gold.



(a) Visual condition: gold border glow on fast hit



(b) Auditory condition: no visual change on hit

Figure 16: Direct comparison of a fast hit in each condition. Under visual feedback (a), the gold glow provides immediate, conspicuous confirmation. Under auditory feedback (b), the interface looks identical regardless of speed—the only feedback is a brief tone. This asymmetry in perceptual salience is central to the speed-chasing effect.

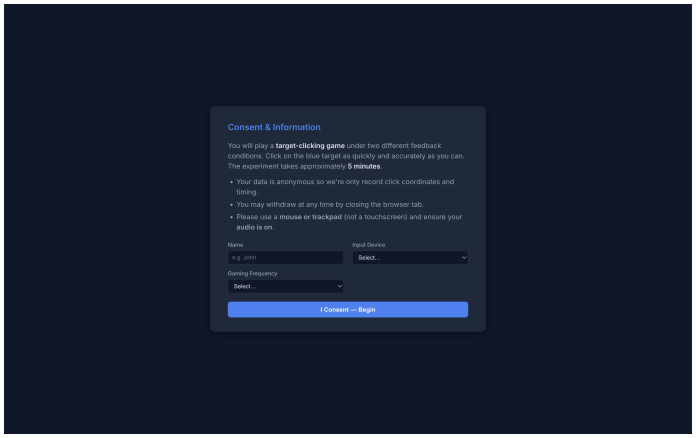


Figure 17: Informed consent and demographic collection screen. Participants provided their name, input device, and gaming frequency before beginning.

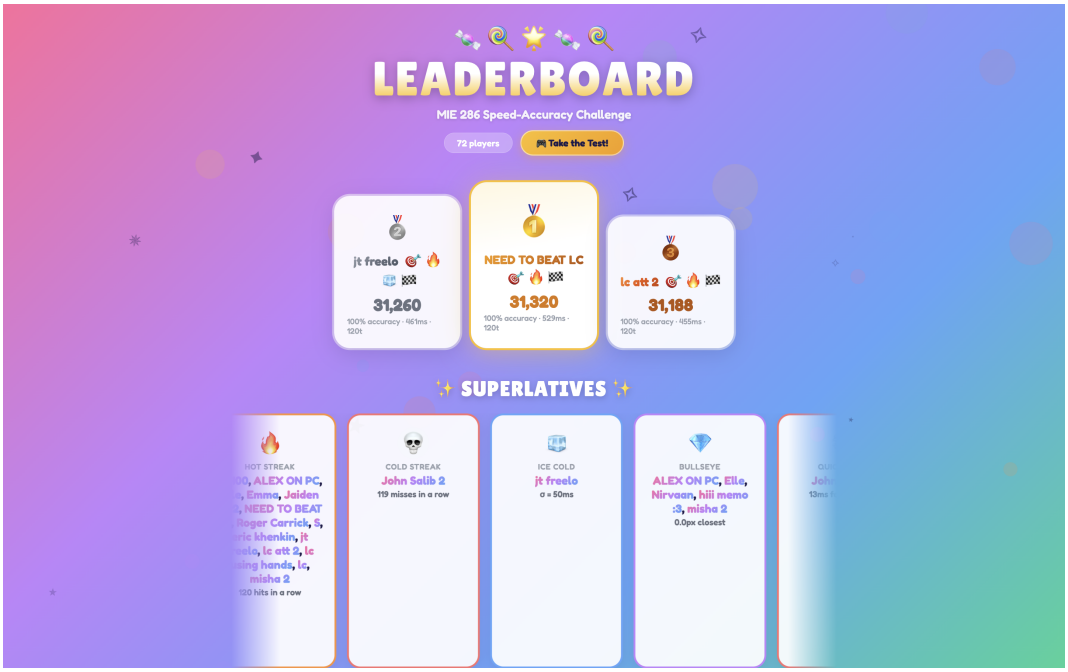


Figure 18: Leaderboard with superlative awards, introduced mid-collection to boost engagement. Post-leaderboard participants were 77 ms faster ($p = .007$), though accuracy was unchanged.

D R Code

All analyses were conducted in R 4.5.3. The pipeline:

1. `01_load_clean.R` — Loads 55 CSVs, excludes RT outliers, aggregates per-participant means.
2. `02_descriptives.R` — Summary tables by condition, device, gaming frequency.
3. `03_assumptions.R` — Shapiro–Wilk tests and QQ plots.
4. `04_inferential.R` — Paired t-test (H_1), Wilcoxon (H_2), multiple regression with interaction (H_3).
5. `05_plots.R` — All figures (spaghetti, density, SAT scatter, learning curve, click scatter, boxplots, covariates, diagnostics).
6. `06_secondary.R` — Tilt effect, order check, variance, block comparison, leaderboard effect, spatial bias, per-participant SAT slopes.

The master script `00_run_all.R` sources all scripts in sequence and is fully reproducible. Full code is in the `analysis/scripts/` directory.